

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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MODEL PRACTICAL EXAMINATION – SET 5

*Answer all questions. Each question carries equal marks (20).
marks: 100*

Total

1. Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as one way, round trip, From, To, Departure, Travelers & Class, Fare Type, International and Domestic Flights, etc.
2. Create a simple cloud software application for Property Buying & Rental process (in Chennai city) using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Buy, Rent, Commercial, Rental Agreement, Area, Place, Range, Property Value, etc.
3. Create a Snapshot of a VM and test it by loading the previous version/cloned VM.
4. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of your present Windows environment.

1. Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as one way, round trip, From, To, Departure, Travelers & Class, Fare Type, International and Domestic Flights, etc.

Flight Reservation System

Aim:

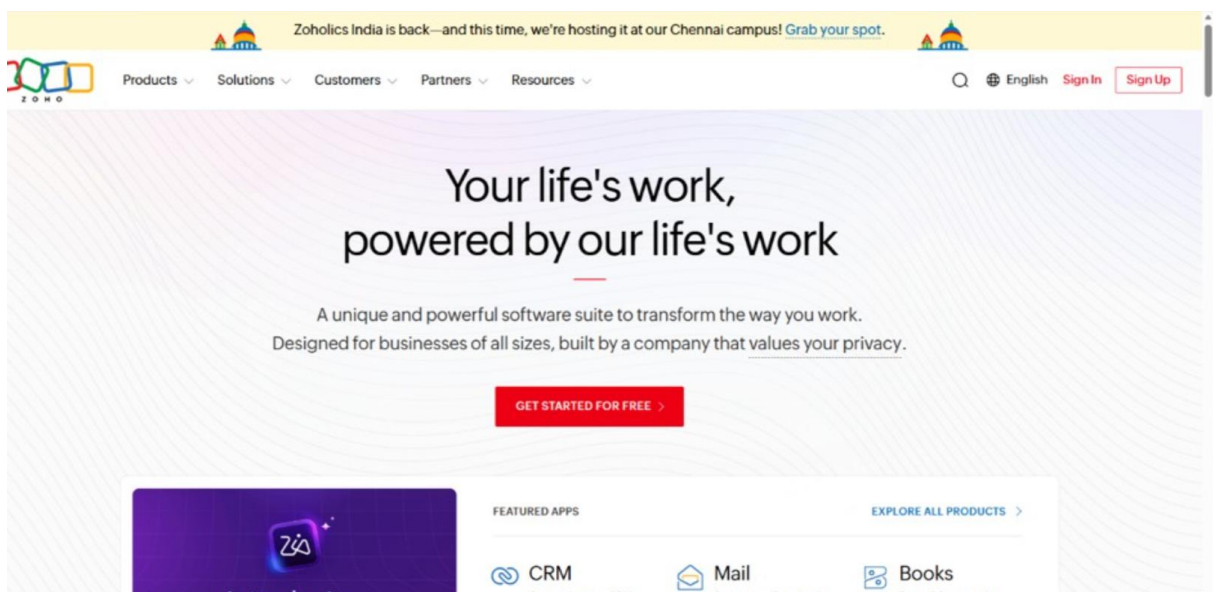
To develop a cloud-based Flight Reservation System using Zoho Creator to demonstrate SaaS.

Steps:

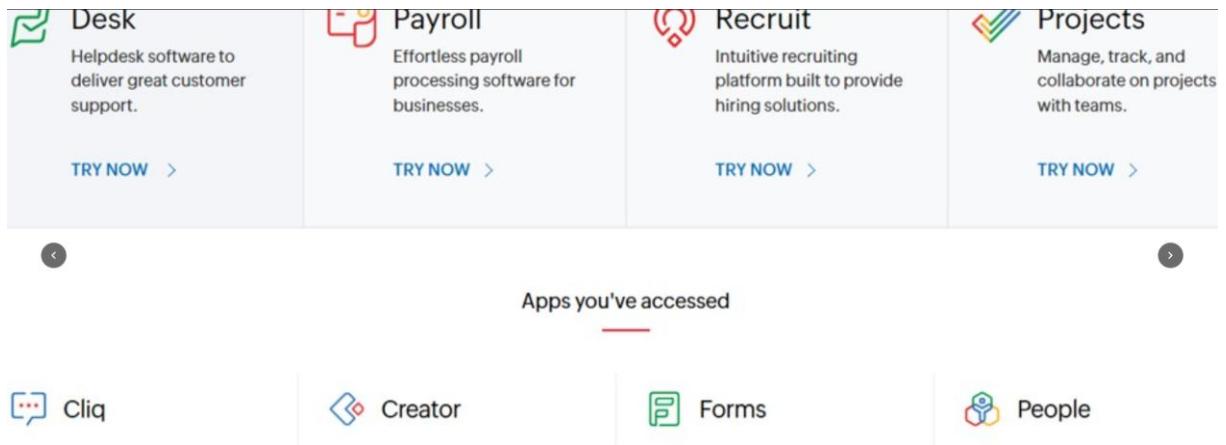
1. Log in to Zoho Creator and click Create New Application.
2. Create a Flight Reservation form.
3. Add fields: From, To, Departure Date, Return Date, Travelers, Class, Fare Type, Flight Type, and Passenger Name.
4. Save the form and create a report to display reservation records.
5. Enter sample data, verify the records, and publish the application.

Implemented:

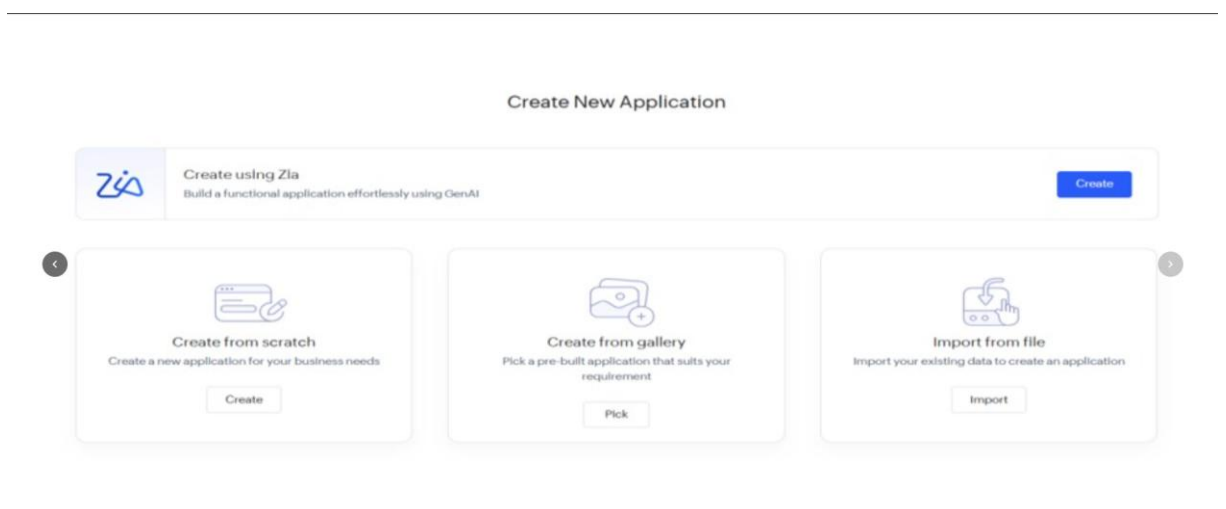
Step 1: Go to zoho.com



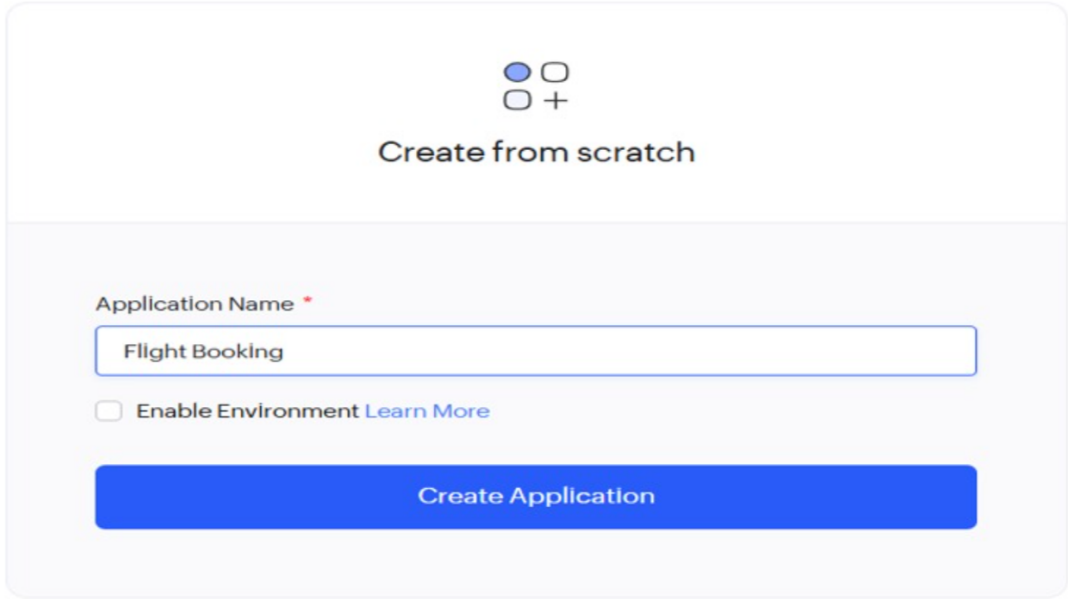
STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch

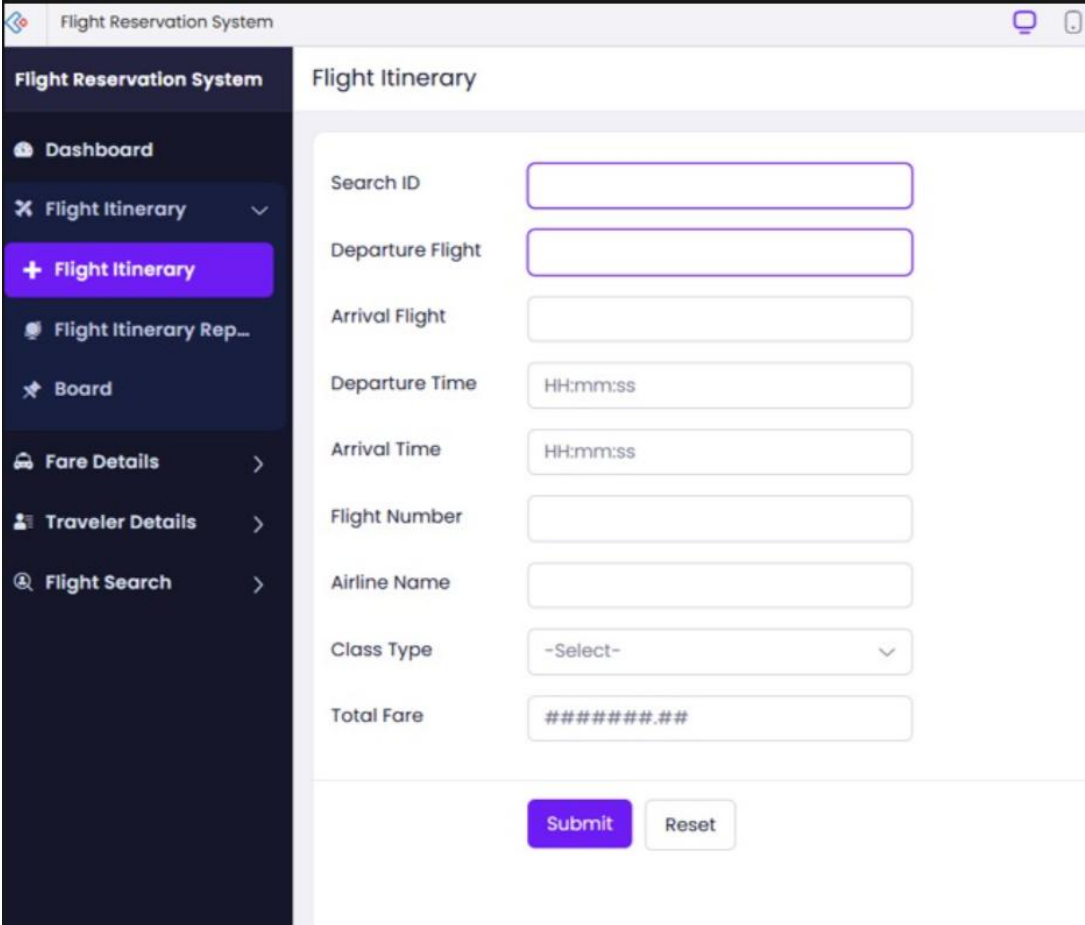


STEP 4: Enter Flight Reservation System as the application name and click Create Application.



The screenshot shows a 'Create from scratch' form. At the top, there is a logo consisting of two blue circles and a plus sign. Below the logo, the text 'Create from scratch' is centered. The form has a light gray background. It contains a text input field labeled 'Application Name *' with the value 'Flight Booking'. Below this, there is a checkbox labeled 'Enable Environment' followed by a link 'Learn More'. At the bottom, there is a large blue button labeled 'Create Application'.

STEP 5: Create a Flight Booking form and add all required fields



The screenshot shows a web application titled 'Flight Reservation System'. The left sidebar contains a menu with the following items: 'Dashboard', 'Flight Itinerary' (expanded), 'Flight Itinerary Rep...', 'Board', 'Fare Details', 'Traveler Details', and 'Flight Search'. The main content area is titled 'Flight Itinerary' and contains a form with the following fields: 'Search ID', 'Departure Flight', 'Arrival Flight', 'Departure Time' (HH:mm:ss), 'Arrival Time' (HH:mm:ss), 'Flight Number', 'Airline Name', 'Class Type' (a dropdown menu with '-Select-' selected), and 'Total Fare' (a text input field with a placeholder '#####.##'). At the bottom of the form, there are two buttons: 'Submit' and 'Reset'.

Flight Reservation System

Flight Reservation System

Dashboard

✕ Flight Itinerary >

🚗 Fare Details >

👤 Traveler Details ▾

✚ Traveler Details

👤 Traveler Details Re...

✈ Board

🔍 Flight Search >

Traveler Details

Itinerary ID

First Name

Last Name

Date Of Birth

dd-MMM-yyyy

Gender

-Select-

Passport Number

Contact Number

🇮🇳 +91 ▾

81234 56789

Email

Traveler Type

-Select-

Submit

Reset

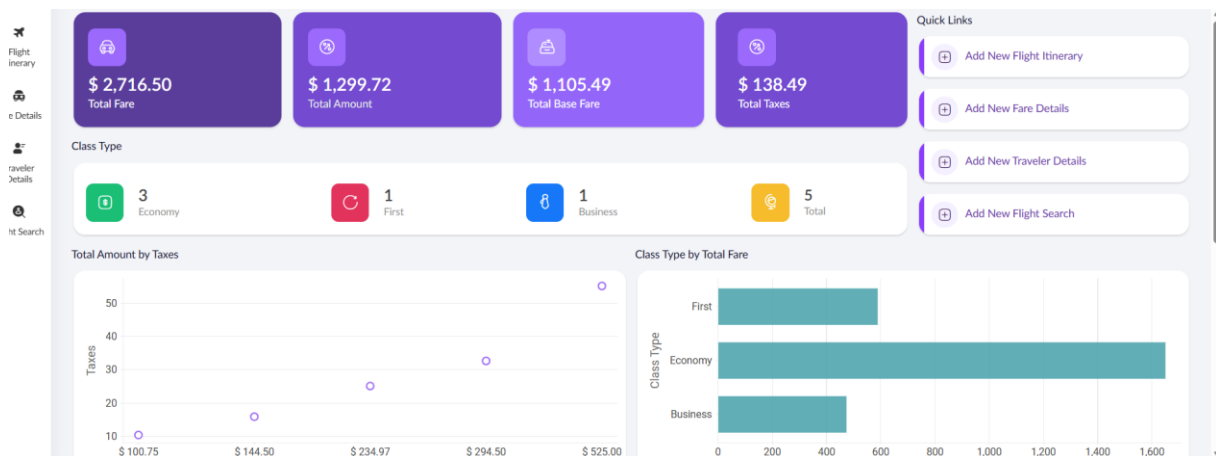
STEP 6: Save the form, enter sample booking records, and view them in the Flight Booking Report.

Traveler Details Report													
		Itinerary ID ▾	First Name ▾	Last Name ▾	Date Of Birth ▾	Gender ▾	Passport Number ▾	Contact Number ▾	Email ▾	Traveler Type ▾			
		2 3	Olivia	Martinez	28-Dec-2001	Female	P99887766	+18106239698	olivia.m@example.com	Child			
		1 2 3	James	Taylor	19-Jun-1995	Male	P55667788	+17639759470	j.taylor@example.com	Senior			
		1 2 3 5	Emma	Williams	03-Jan-2002	Other	P11223344	+17042189136	emma.w@example.com	Senior			
		1	Michael	Chen	22-Aug-1998	Other	P87654321	+18765267315	m.chen@example.com	Child			
		1 3	Sarah	Johnson	15-May-2000	Other	P12345678	+19126059312	sarah.j@example.com	Senior			

Flight Reservation System							
Fare Details Report							
Itinerary ID	Base Fare	Taxes	Service Fee	Total Amount	Currency Code	Fare Description	
1	\$ 199.99	\$ 24.99	\$ 9.99	\$ 234.97	CAD	Economy Flex Fare Option	
2	\$ 450.00	\$ 55.00	\$ 20.00	\$ 525.00	USD	First Class Deluxe Fare	
1	\$ 85.00	\$ 10.25	\$ 5.50	\$ 100.75	GBP	Economy Light Fare Deal	
2							
3							
4							
5							
1	\$ 250.00	\$ 32.50	\$ 12.00	\$ 294.50	EUR	Business Class Premium Fare	
2							
3							
1	\$ 120.50	\$ 15.75	\$ 8.25	\$ 144.50	USD	Economy Class Standard Fare	
2							
3							
4							
5							

Result:

Step 7:The Software has Been Created



2. Create a simple cloud software application for Property Buying & Rental process (in Chennai city) using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Buy, Rent, Commercial, Rental Agreement, Area, Place, Range, Property Value, etc.

Property Buying & Rental System

Aim:

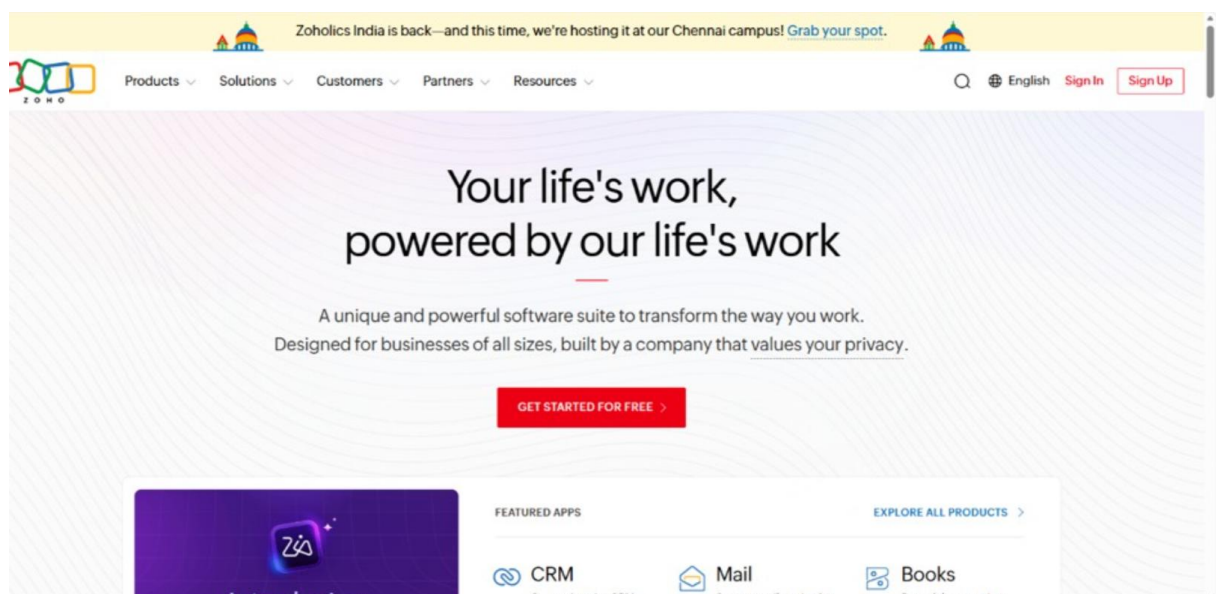
To create a cloud-based Property Buying & Rental System using Zoho Creator.

Steps:

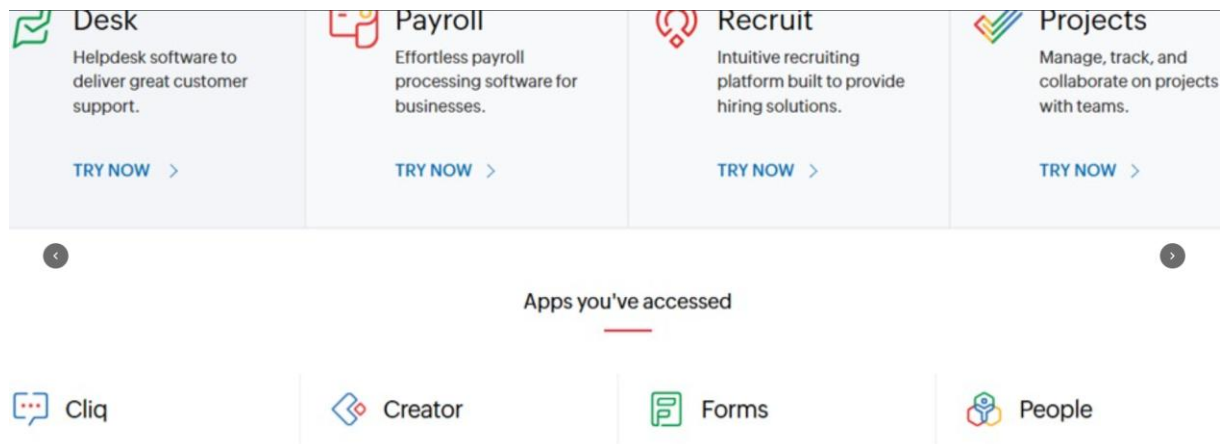
1. Create a new application in Zoho Creator.
2. Create a Property Details form.
3. Add fields: Buy/Rent, Property Type, Area, Location, Price, Rental Agreement, and Owner Details.
4. Create reports to view and search property records.
5. Test the application by entering sample property details.

Implemented

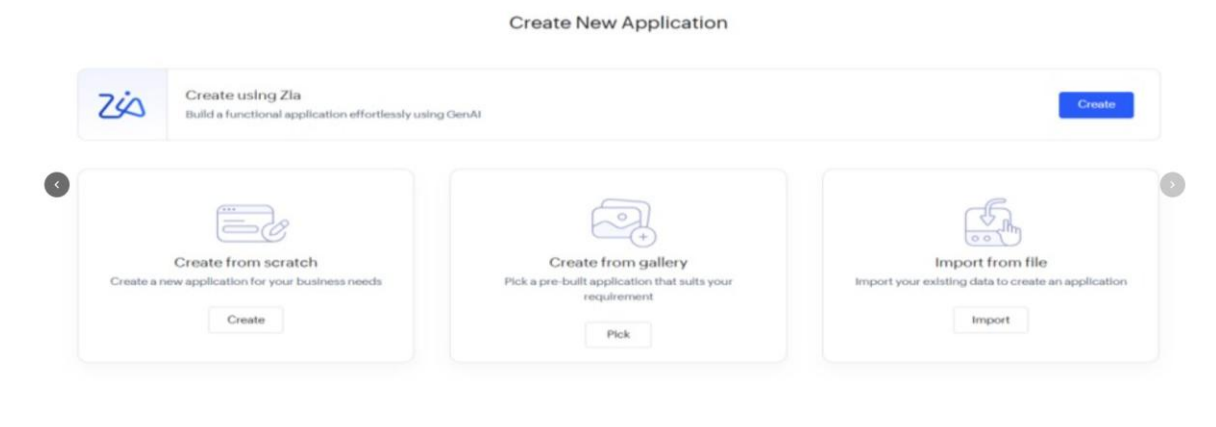
Step 1: Go to zoho.com



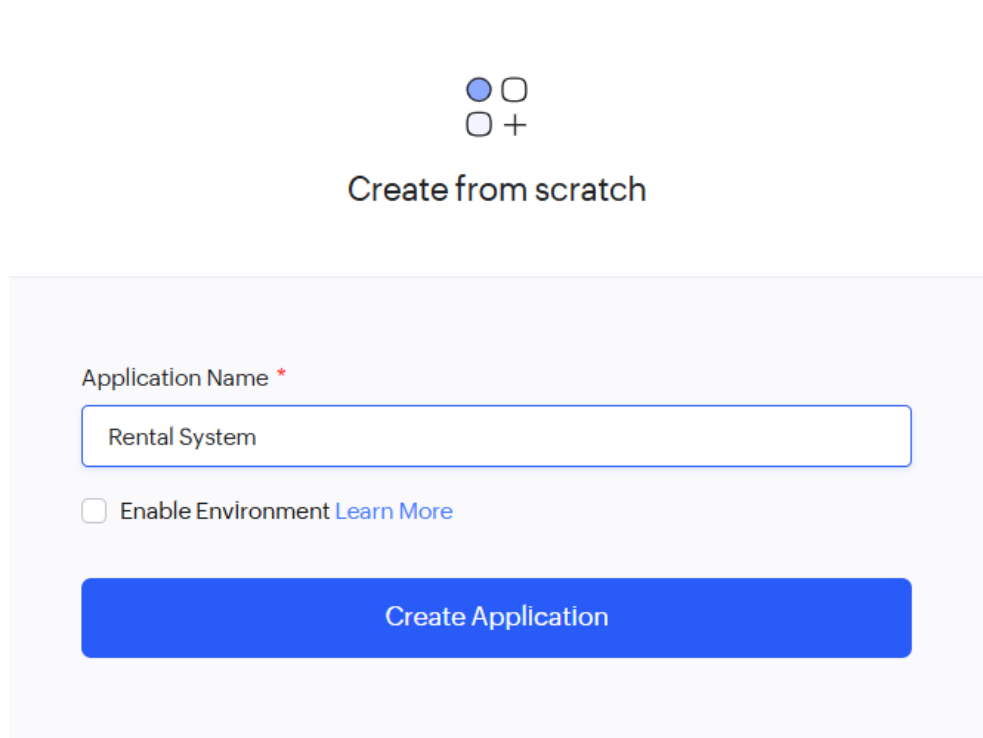
STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch

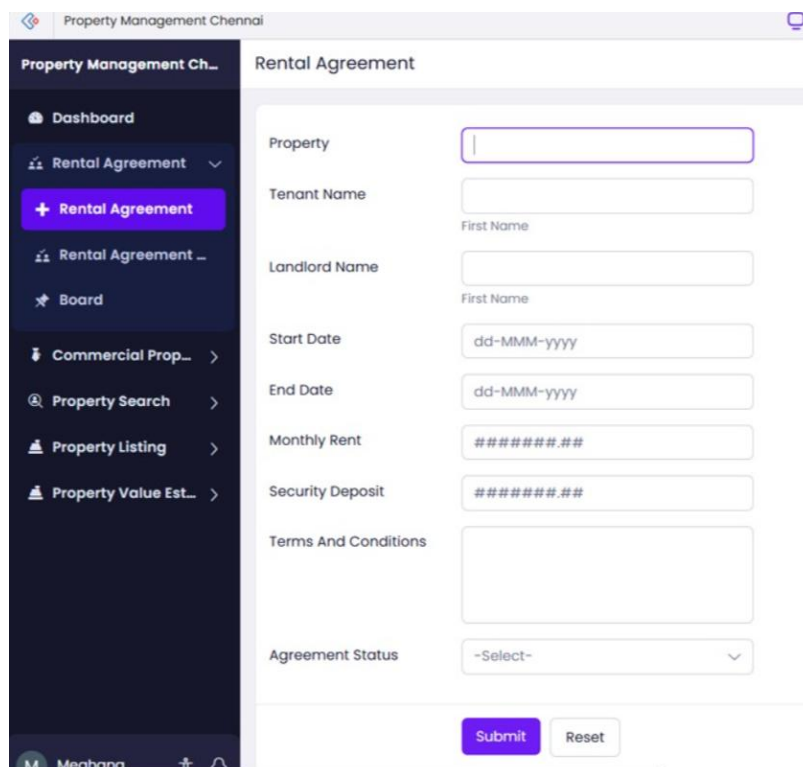


STEP 4: Enter Rental System as the application name and click Create Application.



The screenshot shows a 'Create from scratch' interface. At the top, there are four icons in a 2x2 grid: a blue circle, an empty circle, an empty square, and a plus sign. Below the icons is the text 'Create from scratch'. The main form area has a title 'Application Name *' followed by a text input field containing 'Rental System'. Below this is a checkbox labeled 'Enable Environment' with a link 'Learn More' next to it. At the bottom of the form is a large blue button labeled 'Create Application'.

Step 5: Add fields: Buy/Rent, Property Type, Area, Location, Price, Rental Agreement, and Owner Details.



The screenshot shows the 'Rental Agreement' form within the 'Property Management Chennai' application. The left sidebar contains a menu with items: Dashboard, Rental Agreement (selected), Rental Agreement..., Board, Commercial Prop..., Property Search, Property Listing, and Property Value Est... The main form area is titled 'Rental Agreement' and contains the following fields: Property (text input), Tenant Name (text input with a sub-label 'First Name'), Landlord Name (text input with a sub-label 'First Name'), Start Date (date input with format dd-MMM-yyyy), End Date (date input with format dd-MMM-yyyy), Monthly Rent (text input with format #####.##), Security Deposit (text input with format #####.##), Terms And Conditions (text area), and Agreement Status (dropdown menu with '-Select-' selected). At the bottom of the form are two buttons: 'Submit' and 'Reset'.

Property Management Chennai

Property Management Ch...

Commercial Property

Dashboard

Rental Agreement >

Commercial Prop... >

+ Commercial Prope...

Commercial Prope...

Board

Property Search >

Property Listing >

Property Value Est... >

Property Name

Lease Term

Usage Type

-Select-

Rental Rate

Common Area Charges

Building Type

-Select-

Floor Level

Parking Spaces

Property Status

-Select-

Property

-Select-

Submit Reset

STEP 6: Save the form, enter sample booking records, and view them in the Rental System.

Property Management Chennai

Property Management Ch...

Property Search Report

Dashboard

Rental Agreement >

Commercial Prop... >

Property Search >

+ Property Search

Property Search Re...

Area	Price Range	Property Type	Minimum Area	Maximum Area	Preferred Location	Search Date	User Query
Mylapore	6000000-12000000	Commercial	1500	3000	Mylapore	31-Jul-2026	4 BHK in mature colony
Guindy	4000000-8000000	Rental	1200	2000	Guindy	31-Jul-2026	3 BHK near IT park
OCT	2000000-4000000	Residential	600	1200	OCT	31-Jul-2026	1 BHK close to hospital
Adyar	3000000-6000000	Rental	800	1500	Adyar	31-Jul-2026	2 BHK near sea beach
Anna Nagar	5000000-10000000	Residential	1000	2500	Anna Nagar	31-Jul-2026	3 BHK apartment near metro

Property Management Chennai

Property Management Ch...

Property Listing Report

Dashboard

Rental Agreement >

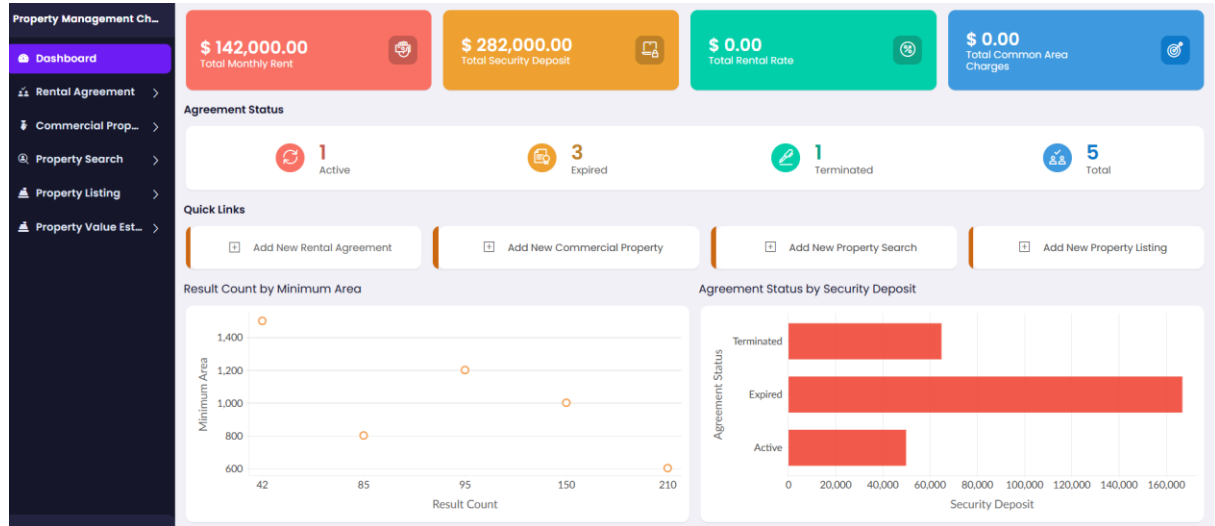
Commercial Prop... >

Property Search >

Property Name	Property Type	Location	Area in Sqft	Price	Property Status	Listing Date	Description	Agent Contact	Property Category
	Residential				Sold			+17544444115	Commercial
	Residential				Sold			+19372025371	Buy
	Rental				Rented			+18018162389	Rent

Result:

Step 7:The Software has Been Created



3. Create a Snapshot of a VM and test it by loading the previous version/cloned VM.

Aim:

To create a snapshot of a virtual machine and restore the previous version using VMware Workstation.

Procedure:

Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.

Step 2: Shut down or suspend the VM and open the Snapshot option.

Step 3: Select Take Snapshot and enter a name such as Before _Changes.

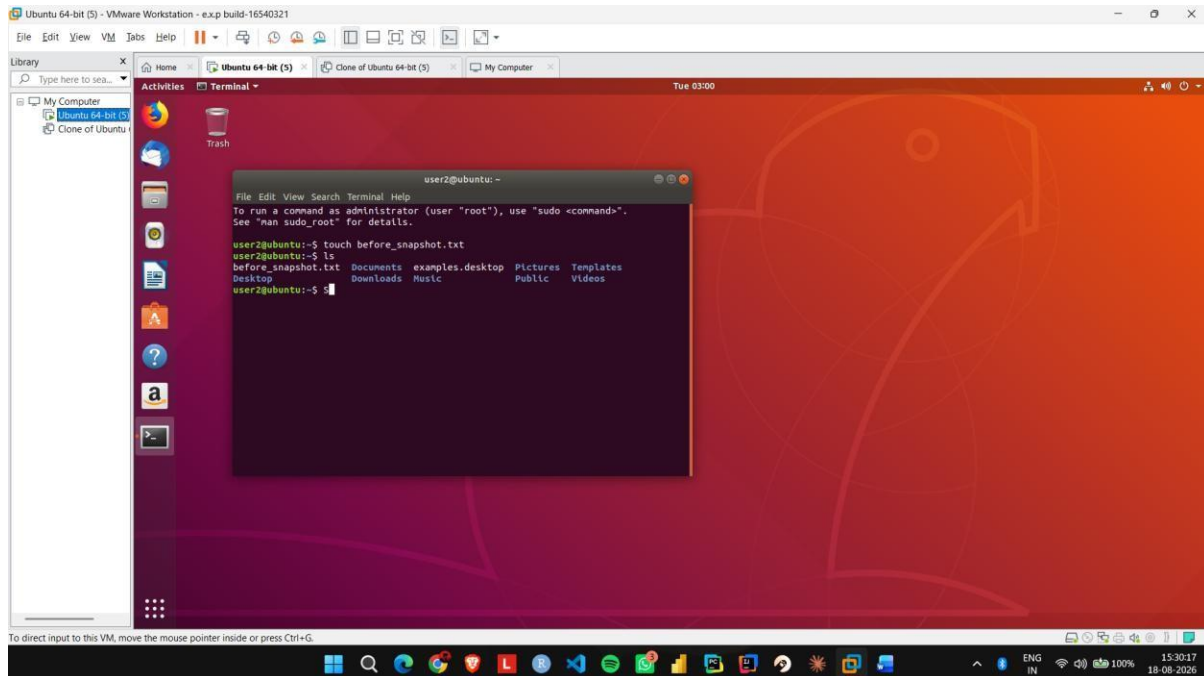
Step 4: Start the VM again and make some changes, such as creating a new file.

Step 5: Open Snapshot Manager, select the Before _Changes snapshot, and click Go To/Restore.

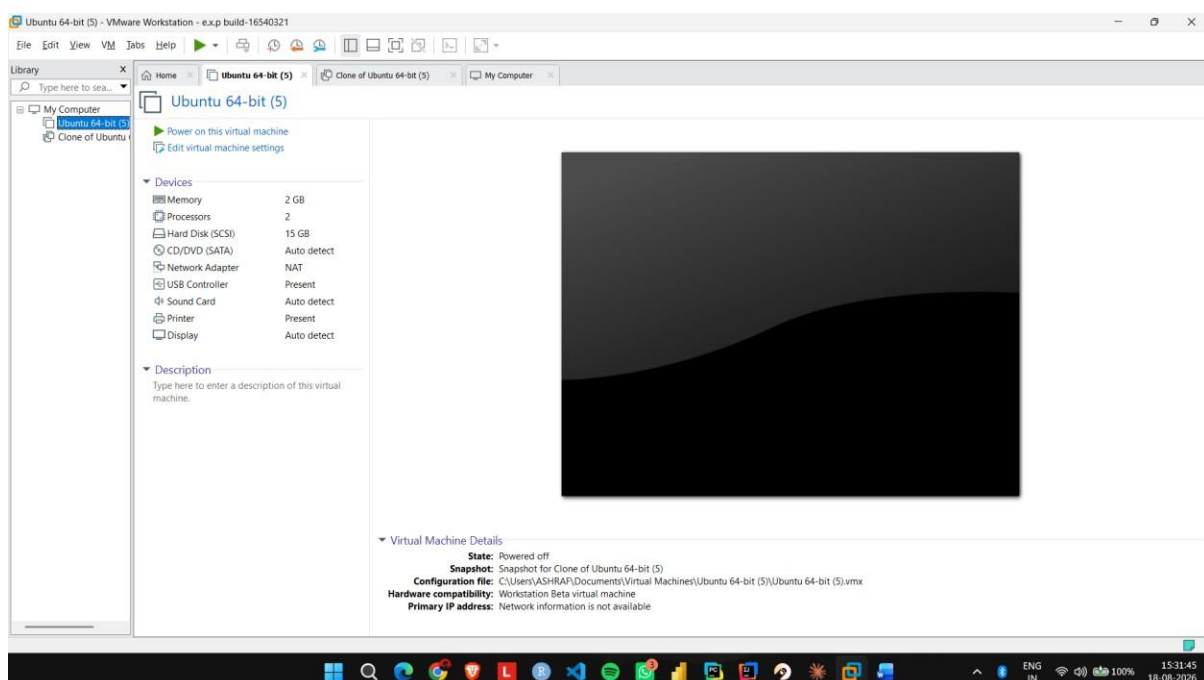
Step 6: Start the VM and verify that the changes made after the snapshot have been removed.

IMPLEMENTATION

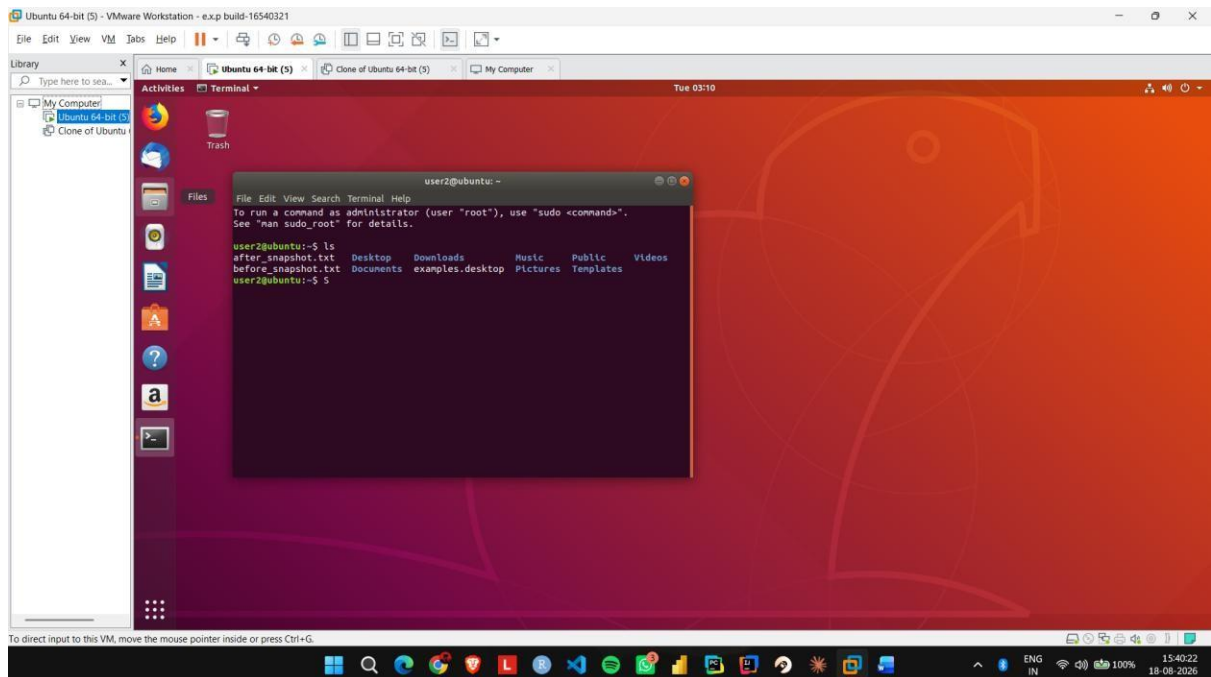
Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.



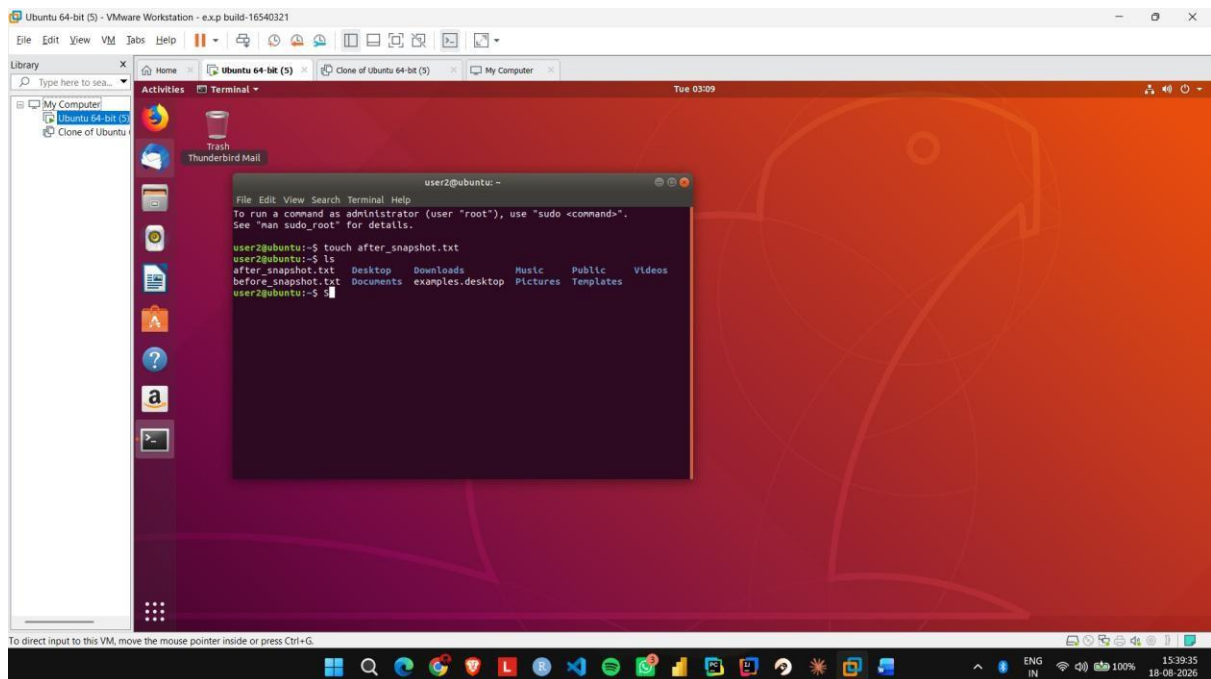
Step 2: Shut down or suspend the VM and open the Snapshot option.



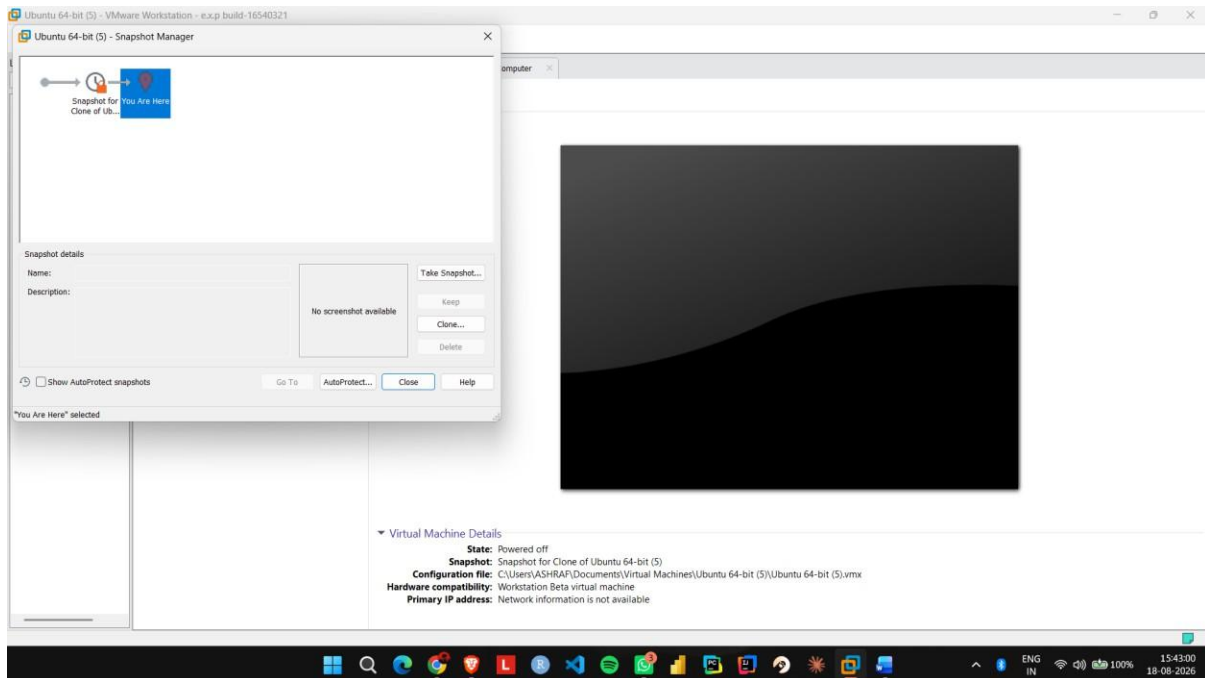
Step 3: Select Take Snapshot and enter a name such as Before Changes.



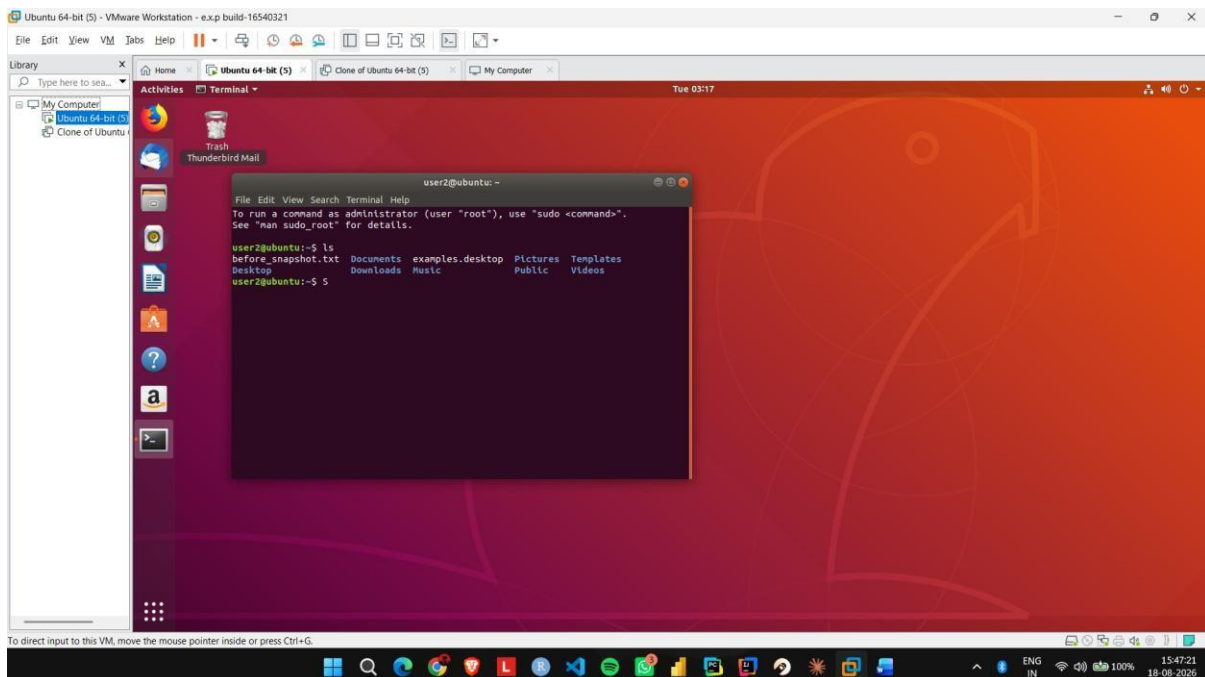
Step 4: Start the VM again and make some changes, such as creating a new file.



Step 5: Open Snapshot Manager, select the Before_Changes snapshot, and click Go To/Restore.



Step 6: Start the VM and verify that the changes made after the snapshot have been removed.



4. Install Virtual box/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.

Aim

To install Virtual Box and create a virtual machine with a different Linux or Windows operating system on the existing Windows environment.

Short Procedure

1. Download and install Oracle Virtual Box on the Windows system.
2. Download the ISO file of the required Linux/Windows OS.
3. Open Virtual Box and select Create New Virtual Machine.
4. Enter the VM name and select the downloaded ISO file.
5. Allocate suitable RAM, CPU cores, and virtual hard-disk space.
6. Start the virtual machine and follow the OS installation instructions.
7. Complete the installation and restart the virtual machine.
8. Verify that the newly installed OS runs successfully inside the virtual environment.

FIG : CREATING VM AND PROVIDING ISO FILE

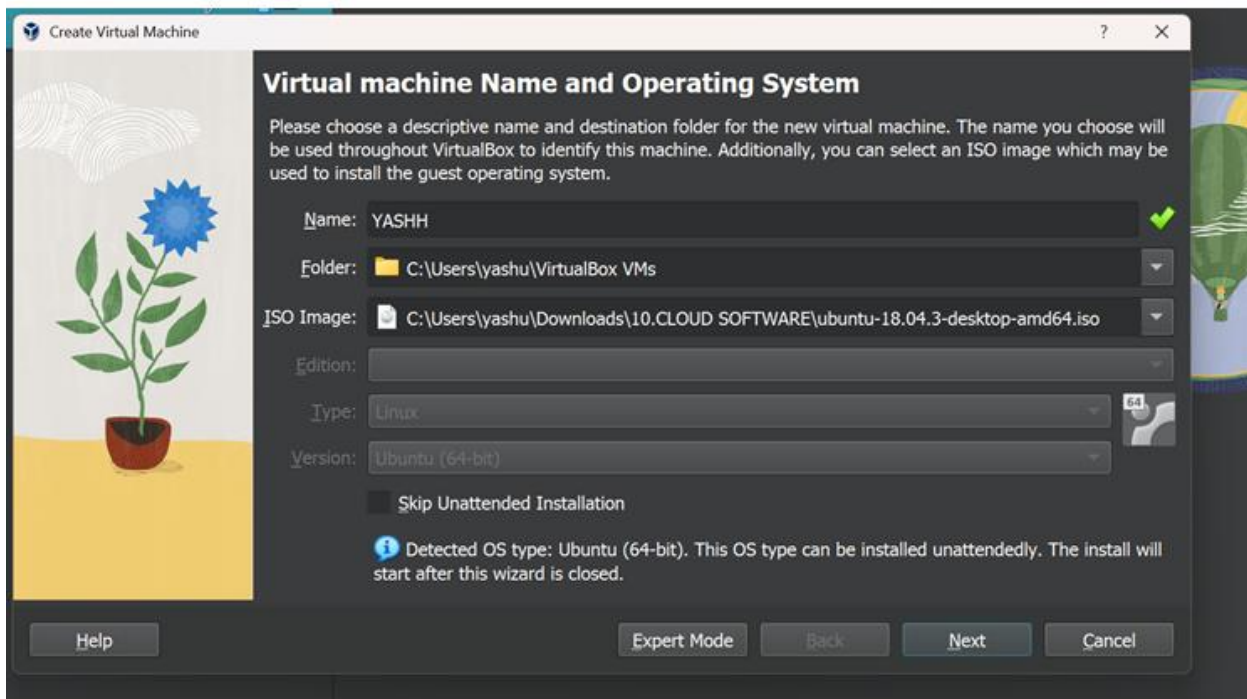


FIG : PROVIDING CREDENTIALS FOR THE VM

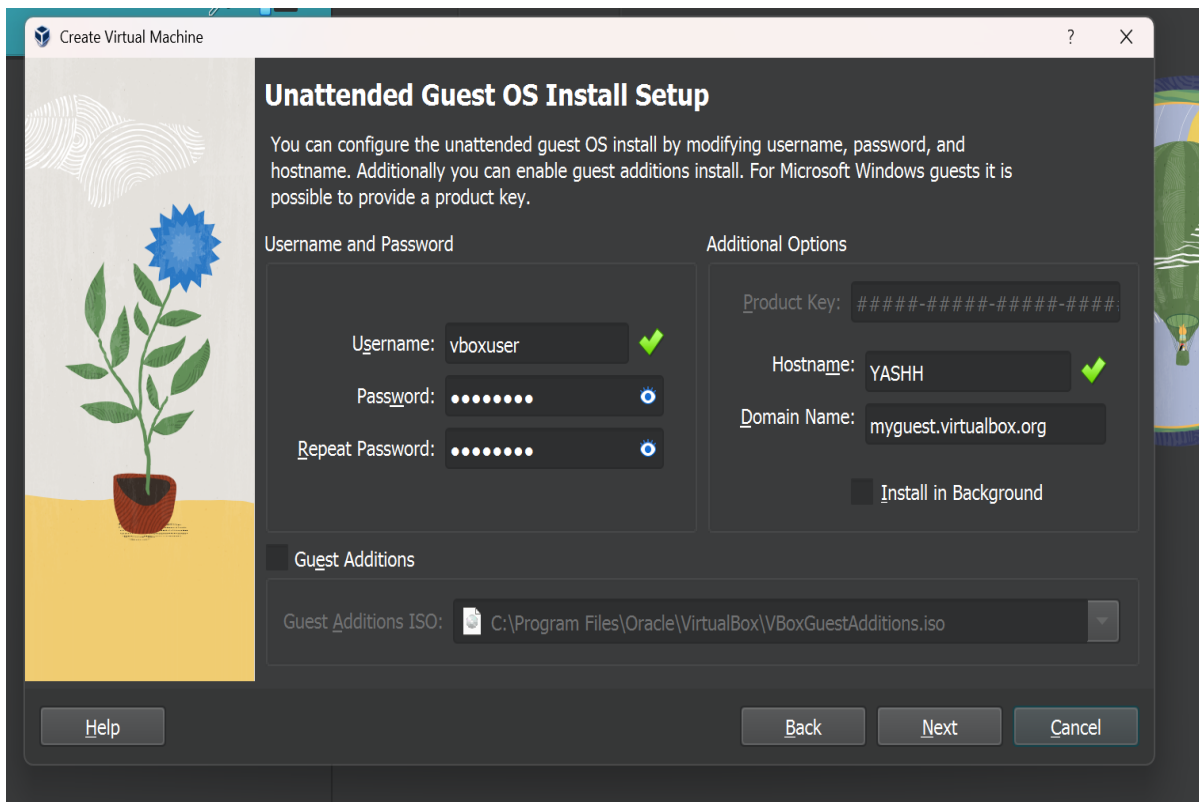


FIG : CREATING VIRTUAL HARD DISK WITH REQUIRED CONFIGURATION

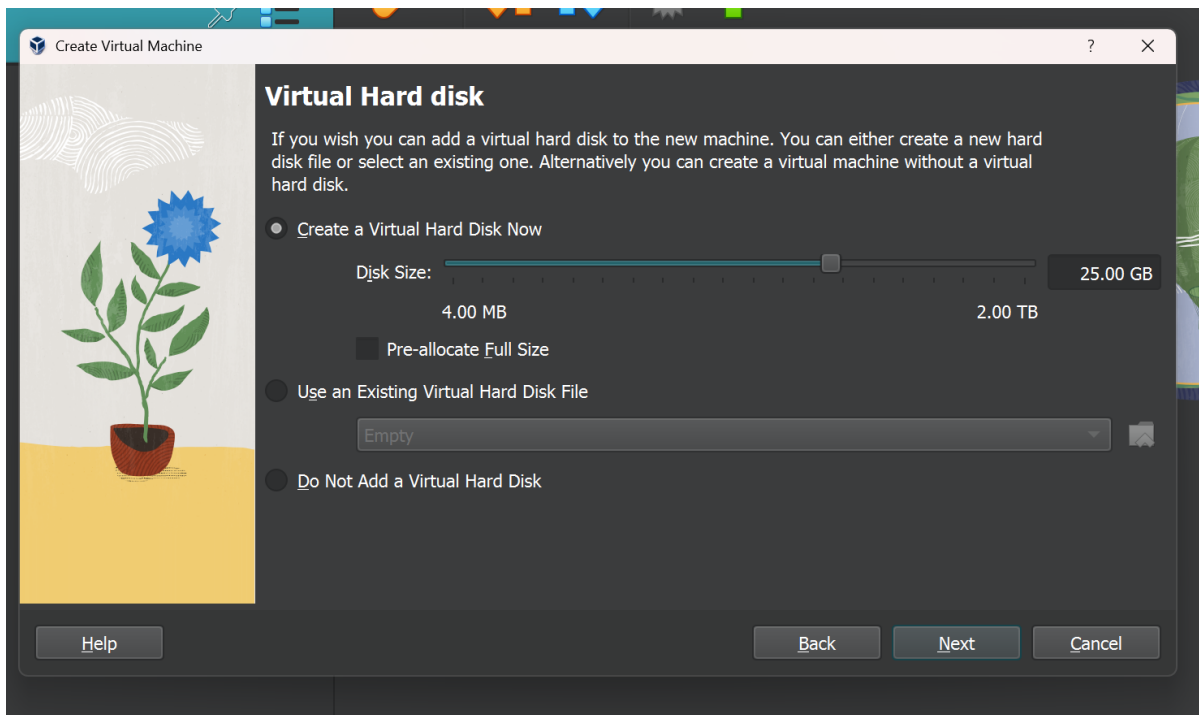


FIG : CONFIGURING THE VM

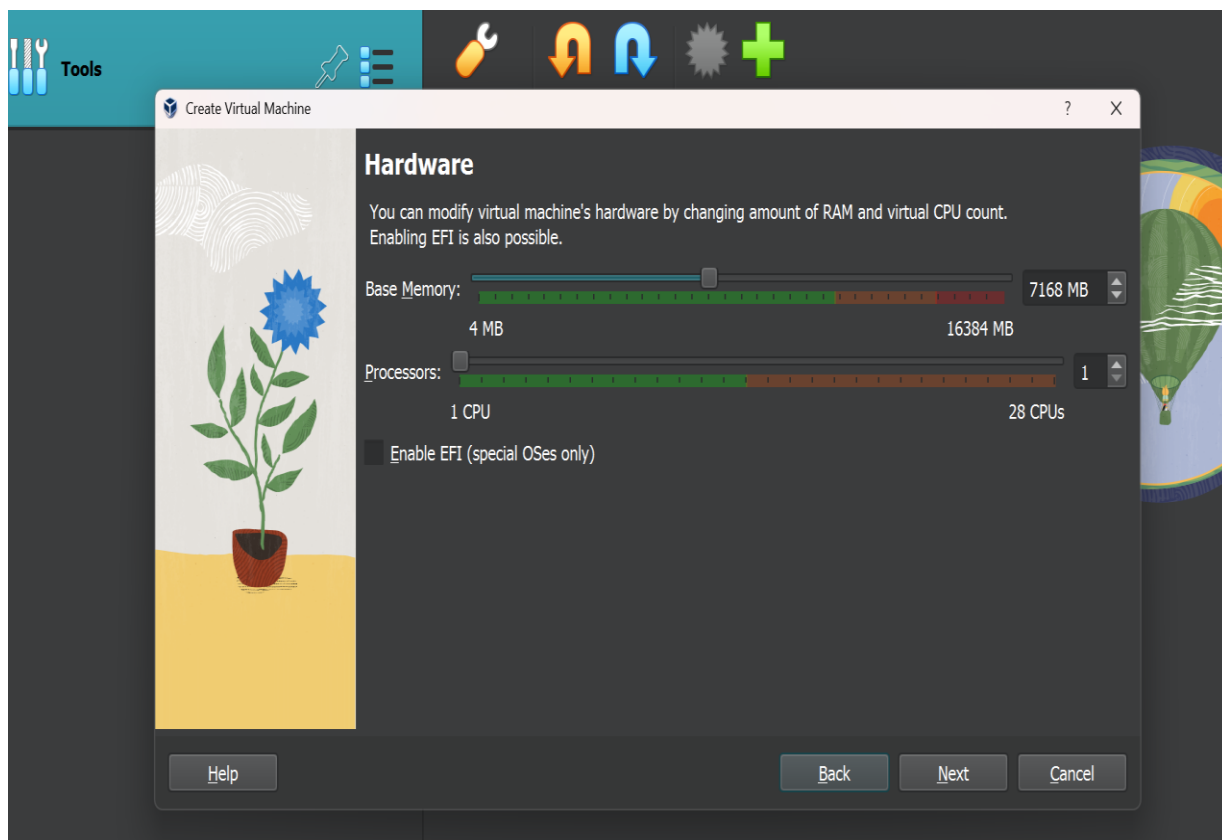


FIG : LAST STEP OF VM

